BUSINESS CASE	
Proposed Project	**Ecosystem Discovery Application** A web-based educational application designed for children in Grades 2–3 (approximately ages 7–8) that introduces freshwater animals through simple, interactive discovery, with an emphasis on basic ecosystem understanding and conservation awareness. All educational content will be written at an approximate Grade 2–3 reading level to ensure clarity and accessibility for the intended audience.
Date Produced	27 January 2026
Background	Environmental education for children in Grades 2–3 is often delivered through textbooks or static digital content, which can limit engagement and make ecosystem concepts difficult to understand. Freshwater ecosystems are among the most familiar and locally relevant environments for young learners, yet there are few age-appropriate, interactive tools that allow children to explore freshwater animals in a simple and meaningful way. This project proposes a web-based educational application focused on freshwater animals commonly found in Saskatchewan or similar North American freshwater ecosystems. Through interactive discovery and visual learning, the application supports early environmental awareness and curiosity-driven learning while remaining aligned with the cognitive level of children aged 7–8. .
Business Need/ Opportunity	There is a need for engaging age-appropriate educational tools that help children in Grades 2–3 actively learn about freshwater ecosystems and conservation. Traditional approaches rely heavily on passive learning, which can reduce understanding and retention at this developmental stage. This project presents an opportunity to enhance early environmental education by providing an interactive, web-based application tailored to the reading and comprehension level of young learners. By focusing on a clearly defined freshwater ecosystem and a limited set of regional animals, the project maintains scope control while delivering meaningful educational value. .
Options	**Option 1: Develop the Ecosystem Educational Discovery Application**

	• Design and implement a web-based educational application • Focus exclusively on freshwater animals • Tailor content and interactions for elementary school children • Follow an MVC architecture with a database-backed model • Deliver a clearly defined Minimum Viable Product (MVP) **Option 2: Do Nothing** • Continue relying on existing static educational resources • No interactive or discovery-based learning experience • No improvement in student engagement or learning outcomes

Cost-Benefit Analysis

Option 1: Develop the Application
Costs:
- Time and effort invested by team members
- Learning and coordination required for design and development
- Potential technical challenges during implementation

Benefits:
- Increased engagement for young learners
- Improved understanding of freshwater ecosystems
- Supports interactive and exploratory learning
- Reinforces conservation awareness at an early age
- Meets course learning objectives related to software engineering, MVC architecture, and project management
- Produces a portfolio-quality academic project

Option 2: Do Nothing

Costs:
- No development or implementation effort required

Benefits:
- Continued reliance on static learning materials may limit student engagement, reduce retention of ecosystem concepts, and provide no opportunity for interactive discovery-based learning.

Recommendation

It is recommended to proceed with **Option 1: Development of the Ecosystem Discovery Application**.
This option provides meaningful educational value by introducing children in Grades 2–3 to freshwater animals through interactive learning at an appropriate reading level. The project remains focused on a clearly defined ecosystem scope, aligns with course requirements, and offers both educational benefits for users and practical software engineering experience for the project team.